Identifying Conics from Their Equations
Answer Key
Algebra 2

Quick Answers

1. -4
2. 144
3. 0
4. 3, -2, 5
5. 3, -4, 1, 5
6. 5, 3, 4
7. -16
8. -5, 2, 4
9. $x = 3$, $y = 9$
10. —

Curriculum

Level: Algebra 2
HSF-IF.A–HSF-IF.C – *problems 5, 9*
HSG-GPE.A.1–A.3 – *problems 1, 2, 3, 4, 6, 7, 8, 10*
Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

8. *If they got 16: stopped at r squared and reported that as the radius*

What this sheet is teaching. Students meet four curves at once and usually try to memorise four pictures. The sheet replaces that with one decision procedure: write the equation in general form, compute $B^2 - 4AC$, and let the sign name the family before any completing-the-square work begins. Only then do the key features come out. Two places to listen carefully: a student who names a circle "ellipse" has skipped the $A = C$ test, and a student who reports 16 as a radius has stopped one square root early.

1. Read the coefficients, then name it. $x^2 + y^2 - 6x + 4y - 12 = 0$

Matching against $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$: there is no xy term, so $B = 0$; the x^2 and y^2 coefficients are both 1.

A	B	C	$B^2 - 4AC$
1	0	1	$0 - 4(1)(1)$

The indicator is $\boxed{-4}$, which is negative, so the curve is an ellipse or a circle. The extra condition decides it: $A = C$ (both are 1), so it is a **circle**.
Why the indicator first? Because completing the square on the wrong assumption costs five lines. Naming the family costs one.

2. Same model, different signs. $9x^2 - 4y^2 = 36$

Move the 36 across: $9x^2 - 4y^2 - 36 = 0$, so $A = 9$, $B = 0$, $C = -4$.

A	B	C	$B^2 - 4AC$
9	0	-4	$0 - 4(9)(-4)$

$B^2 - 4AC = \boxed{144}$, positive, so the curve is a **hyperbola**.
The sign is the whole story. Two squares with *opposite* signs always give $-4AC > 0$. A student who reads "two squares" and says ellipse has looked at the terms instead of at their signs — that is exactly the confusion problem 10 asks about.

3. A missing square. $y = x^2 - 6x + 5$

Rewriting as $x^2 - 6x - y + 5 = 0$: there is an x^2 term but no y^2 term, so $A = 1$, $B = 0$, and $C = 0$.
$B^2 - 4AC = 0 - 4(1)(0) = \boxed{0}$, so the curve is a **parabola**.
The point of the rewrite. $C = 0$ is not "no coefficient" — it is the coefficient zero, and it is what drives the indicator to zero. Every parabola with horizontal or vertical axis has exactly one squared variable, which is why its indicator can only be 0.

4. Circle: find the center and the radius. $x^2 + y^2 - 6x + 4y - 12 = 0$

Solution. Group and complete both squares:

$$(x^2 - 6x) + (y^2 + 4y) = 12$$
$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4$$
$$(x - 3)^2 + (y + 2)^2 = 25$$

Reading the standard form $(x - h)^2 + (y - k)^2 = r^2$ off that line:
$h = -\dfrac{D}{2} = -\dfrac{-6}{2} = \boxed{3}$, $\quad k = -\dfrac{E}{2} = -\dfrac{4}{2} = \boxed{-2}$, $\quad r = \sqrt{h^2 + k^2 + 12} = \sqrt{9 + 4 + 12} = \sqrt{25} = \boxed{5}$.
So the center is $(3, -2)$ and the radius is 5.
Sign trap in the center. The standard form has *minus* h and *minus* k inside the brackets, so $(y+2)^2$ means $k = -2$, not $+2$. Ask the student to check by substituting the center back: the two bracket terms must both vanish.

5. Parabola: vertex, axis, and where it crosses. $y = x^2 - 6x + 5$

Axis of symmetry. With $a = 1$ and $b = -6$, $x = -\dfrac{b}{2a} = -\dfrac{-6}{2} = \boxed{3}$, so the axis is the line $x = 3$.
Vertex y-value. Put $x = 3$ back into the function: $y = 3^2 - 6(3) + 5 = 9 - 18 + 5 = \boxed{-4}$. The vertex is $(3, -4)$.
x-intercepts. Solve $x^2 - 6x + 5 = 0$. It factors as $(x - 1)(x - 5) = 0$, so $x = \boxed{1}$ or $x = \boxed{5}$. Since $a = 1 > 0$ the parabola **opens up**, and the vertex is its lowest point.
A free check. The vertex x must sit exactly halfway between the two intercepts: the midpoint of 1 and 5 is 3. If those two numbers disagree, one of them is wrong, and the student can find out without the answer key.

6. Ellipse: how far out in each direction? $\dfrac{x^2}{25} + \dfrac{y^2}{9} = 1$

Solution. The equation is already in standard form with the denominators $a^2 = 25$ under x^2 and $b^2 = 9$ under y^2:
$a = \sqrt{25} = \boxed{5}$, $\quad b = \sqrt{9} = \boxed{3}$, $\quad c = \sqrt{a^2 - b^2} = \sqrt{25-9} = \sqrt{16} = \boxed{4}$.
Because the larger denominator is under x^2, the major axis is horizontal. So the vertices are $(\pm 5, 0)$, the co-vertices are $(0, \pm 3)$, and the foci are $(\pm 4, 0)$.
The usual slip. Students take a to be the number under x^2 whatever its size. a is defined as the *larger* of the two, and it is the one that goes with the long direction — here the denominators happen to be in the convenient order, which is worth saying out loud so the student does not learn the wrong rule from a friendly example.

7. Two squares, same sign, different sizes. $x^2 + 4y^2 + 2x - 24y + 33 = 0$

Solution. $A = 1$, $B = 0$, $C = 4$, so $B^2 - 4AC = 0 - 4(1)(4) = \boxed{-16}$.
Negative, so it is a circle or an ellipse. Now the extra condition: $A = 1$ but $C = 4$, so $A \neq C$ and the curve is an **ellipse**.
It is not a circle precisely because the two squared terms are scaled differently — the curve is stretched more in one direction than the other. (Completing the square confirms it: $(x+1)^2 + 4(y-3)^2 = 4$, an ellipse centred at $(-1, 3)$.)
What to check in the student's reasoning. The written answer to "why not the other one?" should mention that A and C are unequal — not that "it does not look round".

8. Careful with this one. $x^2 + y^2 + 10x - 4y + 13 = 0$

Solution. Complete both squares:

$$(x^2 + 10x + 25) + (y^2 - 4y + 4) = -13 + 25 + 4$$
$$(x+5)^2 + (y-2)^2 = 16$$

$h = -\dfrac{10}{2} = \boxed{-5}$, $\quad k = -\dfrac{-4}{2} = \boxed{2}$, $\quad r^2 = 16$, $\quad r = \sqrt{16} = \boxed{4}$.
Center $(-5, 2)$, radius 4.
This is the problem the sheet is built around. The right-hand side of the standard form is r^2. A student who writes $r = 16$ has done every hard step correctly and then reported the wrong quantity, and the mistake is invisible unless someone asks "how far is that from the centre?". Have them plot the centre and step 4 units right; 16 units would run off the page.

9. A nonlinear model. $y = -x^2 + 6x$, highest point on the axis $x = 3$.

Solution. Substitute $x = 3$ into the path equation:

$$y = -(3)^2 + 6(3) = -9 + 18 = 9.$$

So the system has the single solution $\boxed{x = 3}$, $\boxed{y = 9}$, and the highest point is $(3, 9)$.

What the coordinates mean. The ball is at its highest when it is 3 metres horizontally from where it was thrown, and at that moment it is 9 metres above the ground.

Why the system has exactly one solution. A vertical line meets a function's graph at most once, and the axis of symmetry is a vertical line — so the system cannot have two answers. That is a structural fact about the parabola's key feature, not an accident of these numbers. Students who solve $-x^2 + 6x = 0$ instead have found the two ground-level points $(0, 0)$ and $(6, 0)$; those are the x-intercepts, not the maximum.

10. Explain the difference. (Open explanation — read the reasoning, not just the verdict.)

The two indicators. For $4x^2 + 9y^2 = 36$: $A = 4$, $C = 9$, so $B^2 - 4AC = -144$, negative with $A \neq C$ — an **ellipse**. For $4x^2 - 9y^2 = 36$: $A = 4$, $C = -9$, so $B^2 - 4AC = +144$ — a **hyperbola**.

A model answer. "Mia is right about the first one. Both equations do have an x^2 and a y^2 term, but that is not what decides the family — the *signs* of those two coefficients are. When both squares are added, the curve closes up into an ellipse. When one is subtracted, no closed curve is possible: as x grows the two terms can stay equal forever, so the graph runs off in two separate branches. The indicator records exactly this: $-4AC$ is negative when A and C agree in sign and positive when they disagree. Mia was looking at which letters are squared instead of at the signs in front of them."

Marking guide. Full credit for any answer that names the *sign difference* between the two squared coefficients, or computes the two indicators and reads them. Partial credit for correctly naming both curves without saying what in the equations made them different. Not sufficient: "one is an ellipse and one is a hyperbola because that is what they look like when you graph them" — true, but it is the picture the sheet is trying to replace with a test she can run on the equation.

Open explanation — $\boxed{\text{open response}}$.